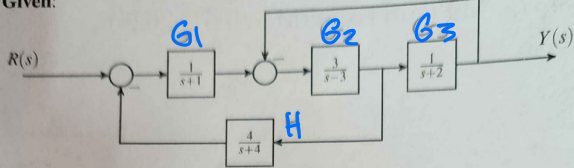


1. Given:



Find:

- 3 ~~X~~ the transfer function, $\frac{Y(s)}{R(s)} = \frac{B(s)}{A(s)}$
 -5 ~~X~~ B. from part A determine the loop gain, L

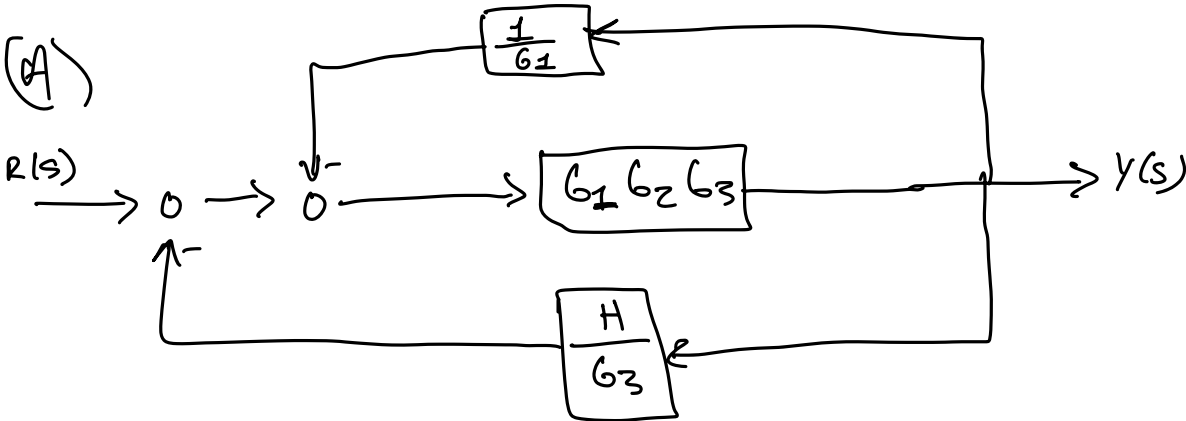
$$G_1 = \frac{1}{s+1}$$

$$G_3 = \frac{1}{s+2}$$

$$G_2 = \frac{2}{s-3}$$

$$H = \frac{4}{s+4}$$

(A)



$$\frac{G_1 G_2 G_3}{1 + \frac{G_1 G_2 G_3}{G_1}} = \frac{G_1 G_2 G_3}{1 + G_2 G_3}$$

$$\frac{\frac{G_1 G_2 G_3}{1 + G_2 G_3}}{1 + \frac{G_1 G_2 G_3}{1 + G_2 G_3} \cdot \frac{H}{G_3}} = \frac{\frac{G_1 G_2 G_3}{1 + G_2 G_3}}{1 + \frac{G_1 G_2 H}{1 + G_2 G_3}}$$

$$= \frac{G_1 G_2 G_3}{1 + G_2 G_3} \cdot \frac{1 + G_2 G_3}{1 + G_2 G_3 + G_1 G_2 H}$$

$$\frac{Y(s)}{R(s)} = \frac{G_1 G_2 G_3}{1 + G_2 G_3 + G_1 G_2 H}$$

$$G_1 = \frac{1}{s+1}$$

$$G_3 = \frac{1}{s+2}$$

$$G_2 = \frac{3}{s-3}$$

$$H = \frac{4}{s+4}$$

$$= \frac{\left(\frac{1}{s+1}\right)\left(\frac{3}{s-3}\right)\left(\frac{1}{s+2}\right)}{1 + \left(\frac{3}{s-3}\right)\left(\frac{1}{s+2}\right) + \left(\frac{1}{s+1}\right)\left(\frac{3}{s-3}\right)\left(\frac{4}{s+4}\right)}$$

$$= \frac{3}{(s+1)(s-3)(s+2)}$$

$$1 + \frac{3}{(s-3)(s+2)} + \frac{12}{(s+1)(s-3)(s+4)}$$

LCD: $(s+1)(s-3)(s+4)(s+2)$
be careful

$$= \frac{3}{(s+1)(s-3)(s+2)} \cdot \frac{(s+1)(s-3)(s+4)(s+2)}{(s+1)(s-3)(s+4)(s+2) + 3(s+1)(s+4) + 12(s+2)}$$

$$= \frac{3}{\cancel{(s+1)}\cancel{(s-3)}\cancel{(s+2)}} \cdot \frac{\cancel{(s+1)}\cancel{(s-3)}(s+4)\cancel{(s+2)}}{(s+1)(s-3)(s+4)(s+2) + 3(s+1)(s+4) + 12(s+2)}$$

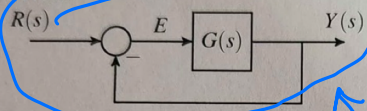
$$= \frac{3(s+4)}{(s+1)(s-3)(s+4)(s+2) + 3(s+1)(s+4) + 12(s+2)} = \frac{3s+12}{s^4 + 4s^3 - 4s^2 - 7s + 12}$$

$$(3) \text{ loop gain} = GS = \frac{\frac{3s+12}{3s+12}}{\frac{s^4+4s^3-4s^2-7s}{3s+12} + \frac{12}{3s+12}} = \frac{1}{\frac{1}{G} + H}$$

$$G = \frac{3s+12}{s^4+4s^3-4s^2-7s} \quad H = \frac{12}{3s+12}$$

$$GH = \frac{12}{s^4+4s^3-4s^2-7s}$$

2. Given:



Negative Feed back

get E from diagram

where $G(s)$ is defined by the time-domain equation: $2\ddot{y}(t) - 2\dot{e}(t) = -4e(t) - 3y(t)$

Find:

~~-2~~ A. the transfer function, $\frac{Y(s)}{R(s)} = \frac{B(s)}{A(s)}$

B. from part A determine the loop gain, L

$$g(t) = 2\ddot{y}(t) - 2\dot{e}(t) = -4e(t) - 3y(t)$$

$$G(s) = 2s^2Y(s) - 2sE(s) = -4E(s) - 3Y(s)$$

$$4E(s) - 2sE(s) = -2s^2Y(s) - 3Y(s) \leftarrow \text{plug in } E(s)$$

$$E = R(s) - Y(s)$$

$$4(R(s) - Y(s)) - 2s(R(s) - Y(s)) = -2s^2Y(s) - 3Y(s)$$

$$4R(s) - 4Y(s) - 2sR(s) + 2sY(s) = -2s^2Y(s) - 3Y(s)$$

$$2s^2Y(s) + 3Y(s) - 4Y(s) + 2sY(s) = 2sR(s) - 4R(s)$$

$$Y(s)(2s^2 + 2s - 1) = R(s)(2s - 4)$$

$$\frac{Y(s)}{R(s)} = \frac{2s - 4}{2s^2 + 2s - 1}$$

$$(B) \quad \frac{\frac{2s-4}{2s-4}}{\frac{2s^2+2s}{2s-4} + \frac{-1}{2s-4}} = \frac{1}{\frac{1}{s} + 1}$$

$$G = \frac{2s-4}{2s^2+2s}, \quad H = \frac{-1}{2s-4}$$

$$GH = \frac{-1}{2s^2+2s}$$

3. Given:

$$\frac{Y(s)}{R(s)} = \frac{s^2 + 37s + 36}{(s+12)(s^2 + 8s + 7)}$$

When the input is a set point,

Find:

A. $y(\infty)$ with the FVT & verify stability with the Routh Array

B. $y(0)$

$$Y(s) = \frac{s^2 + 37s + 36}{s^3 + 20s^2 + 103s + 84} R(s)$$

input is a set point $\Rightarrow$ unit step

$$r(t) = 1, R(s) = \frac{1}{s}$$

(A)

FVT

$$y(\infty) = \lim_{s \rightarrow 0} sY(s) = \lim_{s \rightarrow 0} s \left[\frac{s^2 + 37s + 36}{s^3 + 20s^2 + 103s + 84} \right] \frac{1}{s}$$

$$= \frac{s^2 + 37s + 36}{s^3 + 20s^2 + 103s + 84} \Big|_{s \rightarrow 0} = \frac{36}{84} = \frac{3}{7}$$

Routh Array

s^3	1	103	0
s^2	20	84	0
s^1	98.8	0	0
s^0	84	0	0

$$\frac{20(103) - (84)(1)}{20} = 98.8$$

$$\frac{98.8(84) - 20(0)}{98.8} = 84$$

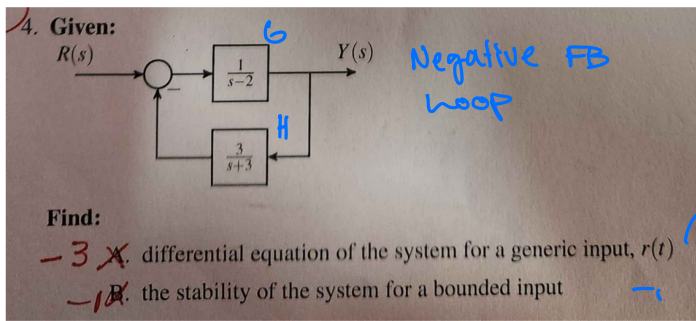
No sign changes,
So system is stable

(B) IVT

$$y(0) = \lim_{s \rightarrow \infty} sY(s) = \lim_{s \rightarrow \infty} s \left[\frac{s^2 + 37s + 16}{s^3 + 20s^2 + 103s + 84} \right] \frac{1}{s}$$

$$= \frac{\frac{s^2}{s^3} + \frac{37s}{s^3} + \frac{16}{s^3}}{\frac{s^3}{s^3} + \frac{20s^2}{s^3} + \frac{103s}{s^3} + \frac{84}{s^3}} \bigg|_{s \rightarrow \infty}$$

$$= \frac{\frac{1}{s} + \frac{37}{s^2} + \frac{16}{s^3}}{1 + \frac{20}{s} + \frac{103}{s^2} + \frac{84}{s^3}} \bigg|_{s \rightarrow \infty} = \frac{0}{1} = \underline{0}$$



① Find $\frac{Y(s)}{R(s)}$

② Get into $Y(s) = R(s)$ form

③ distribute $Y(s)$ and $R(s)$

④ transform to time domain, s^{-1}

$$(A) \quad \frac{Y(s)}{R(s)} = \frac{6}{1+GH} = \frac{\frac{1}{s-2}}{1 + \left(\frac{1}{s-2}\right)\left(\frac{3}{s+3}\right)} = \frac{\frac{1}{s-2}}{1 + \frac{3}{(s-2)(s+3)}}$$

$$= \frac{1}{s-2} \cdot \frac{(s+3)}{(s-2)(s+3)+3} = \frac{s+3}{(s-2)(s+3)+3}$$

$$\frac{Y(s)}{R(s)} = \frac{s+3}{(s-2)(s+3)+3} = \frac{s+3}{s^2+s-3} = \frac{B}{A}$$

$$Y(s)((s-2)(s+3)+3) = (s+3)R(s)$$

$$Y(s)(s^2+s-3) = sR(s)+3R(s)$$

$$s^2 Y(s) + sY(s) - 3Y(s) = sR(s) + 3R(s)$$

$$\ddot{y}(t) + \dot{y}(t) - 3y(t) = \dot{r}(t) + 3r(t)$$

$$\frac{d^2 y(t)}{dt^2} + \frac{dy(t)}{dt} - 3y(t) = \frac{dr(t)}{dt} + 3r(t)$$

$$(B) \quad A(s) = s^2 + s - 3$$

Quadratic formula or Routh Array

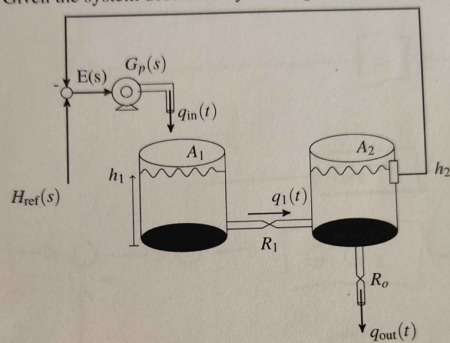
$$\begin{array}{c|ccc} s^2 & 1 & -3 & 0 \\ s^1 & 1 & 0 & 0 \\ s^0 & -3 & 0 & 0 \end{array}$$

↑

$$\frac{1(-3) - 0}{1} = -3$$

sign change, so unstable

5. Given the system described by the diagram and table below



Parameter	Value	Identification
h_2		Height of fluid in tank 2
h_{ref}		Reference height
A_1	2	Area of tank 1
A_2	3	Area of tank 2
R_1	1	Restriction on $q_1(t)$
R_o	1	Restriction on $q_{\text{out}}(t)$
$G_p(s)$	$\frac{1}{s+1}$	Pump

Find: the transfer function $\frac{H_2(s)}{H_{\text{ref}}(s)}$